



5.2 DIODE CHARACTERISTICS

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The most important characteristics when using power diodes is the maximum current in the forward direction (I_{Fmax}), and maximum voltage in the reverse direction (U_{Rmax}).

The important characteristics for a Zener diode are Zener voltage (U_Z), Zener current (I_Z) and maximum dissipation power (P_D).

When working with capacitive diodes it is important to know their maximum and minimum capacitance, as well as values of DC voltage during which these capacitances occur.

With LEDs it is important to know the maximum value of current it is capable of passing. The natural characteristic voltage across a LED depends on the colour and starts at 1.7V for red to more than 2.4v for green and blue.

Current starts at 1mA for a very small glow and goes to about 40mA. High brightness LEDs and “power LEDs” require up to 1 amp and more. You must know the exact current required by the LED you are using as the wrong dropper resistor will allow too much current to flow and the LED will be damaged **instantly**.

The value of this resistors will be covered in another chapter.

Beside universal transistors TUN and TUP (mentioned in Chapter 4.4), there are universal diodes as well. They are marked with DUS (for universal silicon diode) and DUG (for germanium) on circuit diagrams.

DUS = Diode Universal Silicon DUG = Diode Universal Germanium



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